

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 21: Trigonometrical Identities

Previous year board questions, topic-wise: 4 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 The three basic identities

- Assertion (A): If $\sin^2 A + \sin A = 1$, then $\cos^4 A + \cos^2 A = 1$. Reason (R): $1 - \sin^2 A = \cos^2 A$. (a) A true, R false (b) A false, R true (c) both true, and R is the correct reason for A (d) both true, but R is not the correct reason for A **ICSE 2025, 1 mark**
- Statement 1: $\sin^2 \theta + \cos^2 \theta = 1$. Statement 2: $\operatorname{cosec}^2 \theta + \cot^2 \theta = 1$. Which is valid? (a) only 1 (b) only 2 (c) both (d) neither **ICSE 2024, 1 mark**

2 Proving identities

- Prove that $\frac{(\cot A + \tan A - 1)(\sin A + \cos A)}{\sin^3 A + \cos^3 A} = \sec A \operatorname{cosec} A$. **ICSE 2025, 4 marks**
- Factorise $\sin^3 \theta + \cos^3 \theta$. Hence prove that $\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} + \sin \theta \cos \theta = 1$. **ICSE 2024, 4 marks**

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